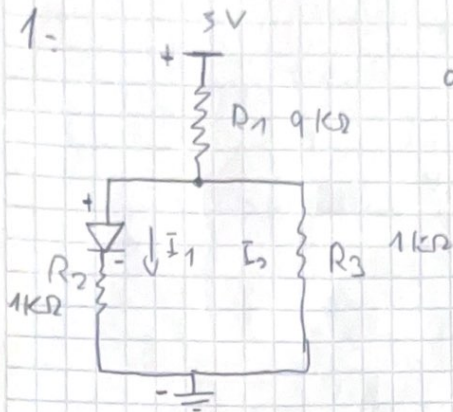
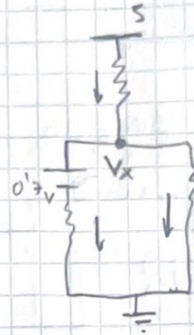


1-



a) $V_D = 0.7$

este apagado



$$\frac{5 - V_x}{9} = \frac{V_x - 0.7}{1} + \frac{V_x - 0}{1}$$

$$5 - V_x = 9V_x - 6.3 + 9V_x$$

$$\frac{11.3}{19} = V_x$$

$V_x = 0.59 < 0.7$ esta apagado así que

$$I_{D1} = 0 \text{ A}$$

$$I_{D2} = \frac{5 \text{ V}}{10 \text{ k}\Omega} = 0.5 \text{ mA}$$

b) Ideal

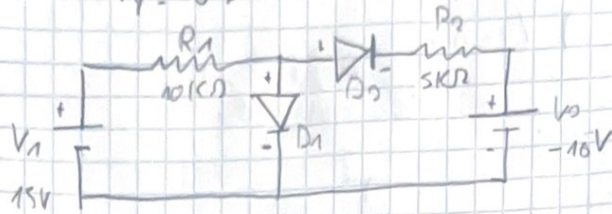


$$\frac{5 - V_x}{9} = \frac{V_x}{1} + \frac{V_x}{1}$$

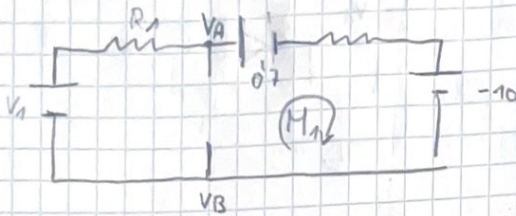
$$V_x = \frac{5}{19}$$

$$I_1 = \frac{\frac{5}{19}}{1000} = 0.26 \text{ mA} = I_2$$

2- $V_Y = 0.7$



Hipotesis D_1 apagado



$$15 - 0.7 - (-10) - R_1 I_1 - R_2 I_1 = 0$$

$$24.3 - 15 I_1 = 0 \quad I_1 = 1.62 \text{ mA}$$

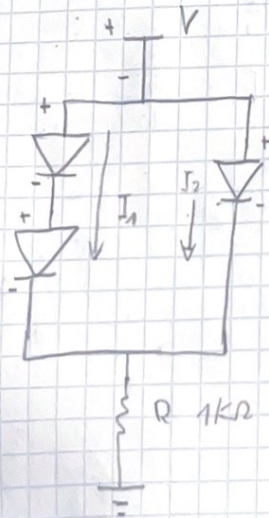
$$I_{D2} = 1.62$$

$$I_{D1} = 0$$

$$V_{D2} = 0.7 \text{ V}$$

$$V_{D1} = 15 - 1.62 \cdot 10 = -16 \text{ V}$$

3-

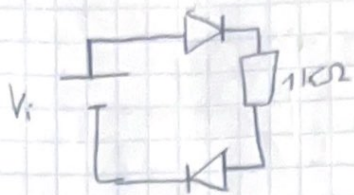


$$\frac{V_x - 0.7}{1000} = 4.3 \text{ mA}$$

$$I_1 = 0 \text{ mA}$$

4: $V_y = 0.7V$

$V_{R1}?$



$V_i = 1V$

$I_R = 0V$

$V_i < V_y$ no pasa corriente

$V_R = 0$

$V_i = 0V$

$I_R = 0V$

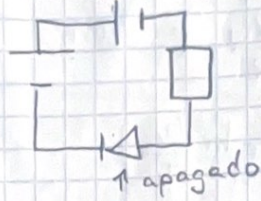
$V_i < V_y$ no pasa corriente

$V_R = 0$

$V_i = 1V$

$V_i > V_y$ pasa corriente

$0.3 < 0.7$ no pasa



$1 - 0.7 = 0.3V$

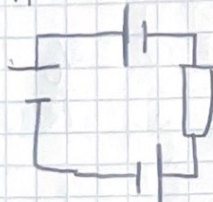


$I_R = 0V$

$V_R = 0$

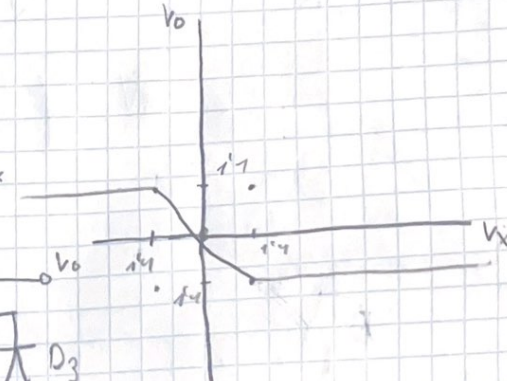
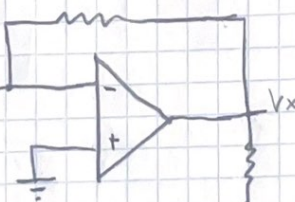
$V_i = 2$

$I_R = \frac{2 - 0.7 - 0.7}{1000} = 0.6mA$



$V_R = 0.6V$

$S = V_i$



$\frac{V_i - V_o}{R} = \frac{-V_x}{R}$

$V_x = -V_i$

$I = \frac{-V_i - V_o}{R}$

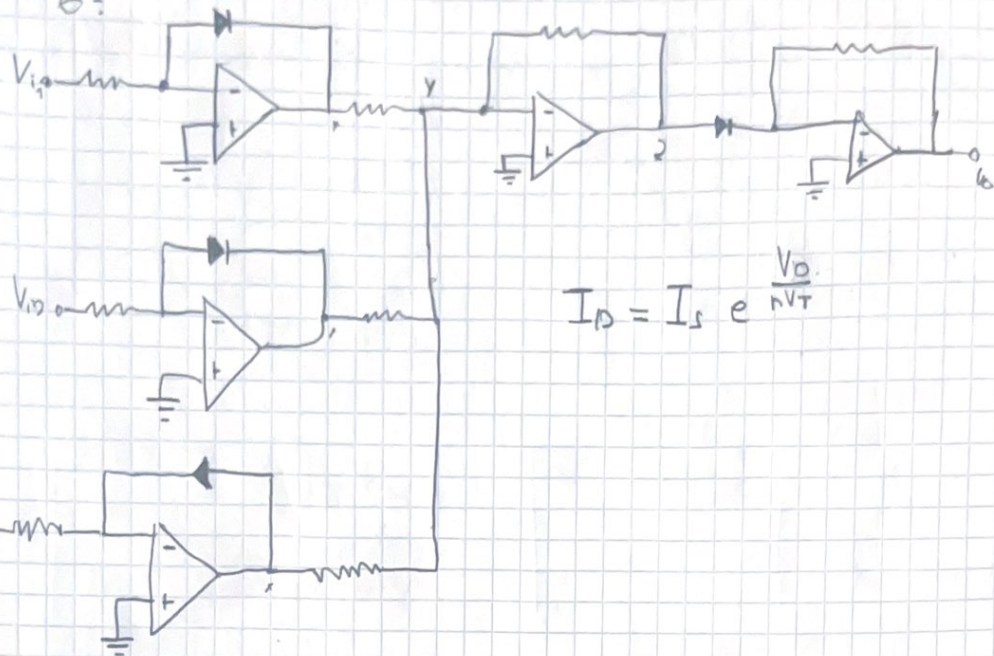
$\frac{-V_i - V_o}{R} = -1.4 + 1.4 \quad R = 1$

$V_o = -V_i \quad S: V_i > 1.4 \quad V_o = -1.4$

$S: V_i < -1.4 \quad V_o = 1.4$

$S: V_i = 0 \quad V_o = 0$

* 6:



$$I_D = I_S e^{\frac{V_D}{nV_T}}$$

$$I_r = I_D$$

$$\frac{V_i}{R} = I_S e^{\frac{V_D}{nV_T}}$$

$$V_{x1} = -nV_T \ln \left(\frac{V_{i1}}{I_S R} \right)$$

$$V_{x2} = -nV_T \ln \left(\frac{V_{i2}}{I_S R} \right)$$

$$V_{x3} = nV_T \ln \left(\frac{1}{I_S R} \right)$$

$$\frac{V_{x1}}{R} + \frac{V_{x2}}{R} + \frac{V_{x3}}{R} = -\frac{V_2}{R}$$

$$V_2 = nV_T \ln \left(\frac{V_{i1}}{I_S R} \right) + nV_T \ln \left(\frac{V_{i2}}{I_S R} \right) - nV_T \ln \left(\frac{1}{I_S R} \right)$$

$$\ln(a) + \ln(b) = \ln(ab)$$

$$V_2 = nV_T \ln \left(\frac{V_{i1} V_{i2}}{I_S R} \right)$$

$$V_0 = -RI_S e^{\left(\frac{nV_T \ln \left(\frac{V_{i1} V_{i2}}{I_S R} \right)}{nV_T} \right)}$$

$$I_D = \frac{0 - V_0}{R}$$

$$I_D = I_S e^{\frac{V_2}{nV_T}}$$

$$V_0 = -RI_S \cdot \frac{V_{i1} V_{i2}}{I_S R} \Rightarrow V_0 = -V_{i1} V_{i2}$$

$$V_0 = -RI_S e^{\frac{V_2}{nV_T}}$$



	Procesos NMOS	CMOS PMOS
$I_{min}(mA)$	50	50
$V_{th}(mV)$	500	500
$V_{DD}(V)$	1	1
$R_{on}(k\Omega)$	$35(I_n/W_n)$	$70(I_p/W_p)$
$C_{ox}(fF)$	$25(I_n \cdot W_n)$	$25(I_p \cdot W_p)$

1- $W_n = 4 W_{min}$ $I_n = I_{min}$

$$R_n = \frac{R_{on} \cdot I_n}{W_n} = \frac{R_{on} I_{min}}{4 W_{min}} = 0.875 k\Omega$$

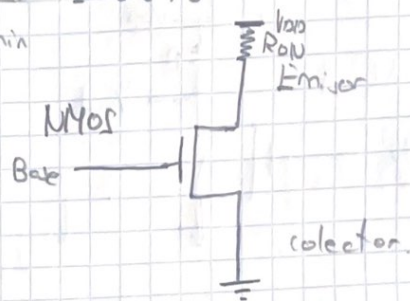
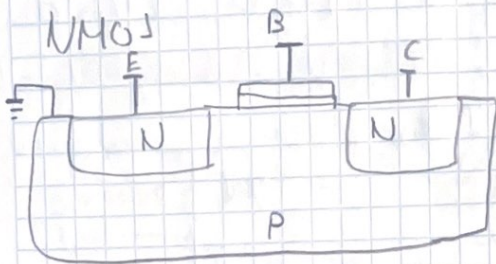


Table verdad

Entree	Salida
0	0
1	1

2- PMOS



$$R_{on} = 70 \cdot \frac{I_p}{W_p}$$

$$\frac{10}{70} = \frac{I_p}{W_p} \Rightarrow \frac{W_p}{I_p} \leq 7$$

by probando

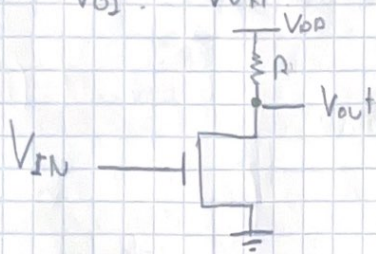
$$\frac{500}{50} \leq 7 \quad \text{falso}$$

$$\frac{500}{2 \cdot 50} \leq 7 \quad \text{verdad}$$

así que $W_p = W_{min}$

$$I_p = 2 \text{ } \mu\text{min.}$$

3- V_{ol} ? V_{OH} ?



$$V_{OH} = V_{DD}$$

High

Transistor OFF

$$I = 0; I \cdot R = 0$$

$$V_{DD} - V_O = 0, \quad V_{DD} = V_O$$

$$V_{OL} = I R_N$$

$$I = \frac{V_{DD}}{R + R_N}$$

$$R_N = 35 \cdot \frac{50}{500} = 3.5 \text{ k}\Omega$$

$$V_{OL} = \frac{V_{DD} R_N}{R + R_N} \leq 0.2 \text{ V}$$

$$R_N \leq \frac{0.2 R}{0.8} = \frac{0.2}{0.8} R$$

Pruebo con 0.1:

$$\frac{R_N}{R + R_N} \leq 0.1$$

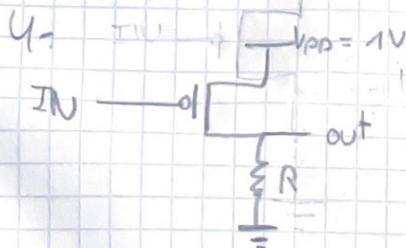
$$0.9 R_N \leq 0.1 R$$

$$R_N \leq 1.11 \text{ k}\Omega$$

$$\frac{R_N}{R + R_N} \leq 0.1; \quad 0.9 R_N \leq 0.1 R$$

$$R_N = 35 \cdot \frac{L_n}{W_n} \leq 1.11$$

35 $\neq$ 1.11 con los de mínimo no se puede



$$5- NM_L = V_{iL} - V_{oL} = 1'6V$$

$$NM_H = V_{oH} - V_{iH} = 2'5V$$

V_{oH} = nivel alto de salida

V_{oL} = nivel bajo de salida

V_{iL} = valor máximo de entrada interpretado por el inversor como '0'

V_{iH} = valor mínimo de entrada interpretado por el inversor como '1'

$$6- C_L = 10 \text{ pF} = 10^{-12} \text{ F}$$

$$a) t_{PLH} = t_{PHL} = 0'69 RC_L$$

$$\downarrow \qquad \qquad \downarrow$$

$$0'69 R_P C_L \quad 0'69 R_N C_L$$

$$R_P = 70 \frac{I_P}{W_P} = 70 \cdot \frac{50}{500} = 7 \text{ k}\Omega$$

$$R_N = 35 \frac{I_N}{W_N} = 35 \cdot \frac{50}{500} = 3'5 \text{ k}\Omega$$

$$t_{PLH} = 0'69 \cdot 7000 \cdot 10^{-12} = 48'3 \text{ ns}$$

$$t_{PHL} = 0'69 \cdot 3500 \cdot 10^{-12} = 24'13 \text{ ns}$$

$$\downarrow$$

$$1 \cdot 10^{-11}$$

$$b) \left. \begin{array}{l} 0'69 R_P C_L \rightarrow R_P = 7 \text{ k}\Omega \\ 0'69 R_N C_L \rightarrow R_N = 3'5 \text{ k}\Omega \end{array} \right\} t_{PLH} = t_{PHL} \rightarrow R_P = 2 R_N$$

$$I_P = I_N = I_{min}$$

$$W_P = W_{min}$$

$$W_N = 2 W_{min}$$

$$\frac{W_P}{I_P} = 2 \frac{W_N}{I_N}$$

$$c) \quad t_p = t_{pLH} = t_{pHL} = 2 \text{ ns}$$

$$t_{pLH} = 0.69 R_p C_L = 2 \text{ ns}$$

$$0.69 \cdot 10 \cdot 10^{-12} R_p = 2 \cdot 10^{-9}$$

$$R_p = 289.855 \, \Omega$$

$$R_p = 70 \frac{I_p}{I_{wp}}$$

$$0.289855 = 70 \cdot \frac{50}{I_{wp}}$$

$$I_p = I_{min}$$

$$I_{wp} = 24.15 \text{ Wmin}$$

$$I_{wp} = 24.15 \text{ Wmin}$$

Segun antes

$$I_N = I_{min}$$

$$W_N = 12.075 \text{ Wmin}$$

d) N° etapas

$$N = \ln \left(\frac{C_1}{C_{T1}} \right) = h \left(\frac{10 \cdot 10^{-12}}{2.8125 \cdot 10^{-15}} \right) = 8.176$$

$$C_{TN} = 1.5 \cdot (C_{xw} + C_{xp}) = 2.8125 \quad C_{01} = 1.875$$

$$C_{xw} = 25 \cdot 0.05 \cdot 2 \cdot 0.5 = 1.25$$

$$C_{xp} = 25 \cdot 0.05 \cdot 0.5 = 0.625$$

$$S = \sqrt[8]{\frac{10 \cdot 10^{-12}}{2.8125 \cdot 10^{-15}}} = 2.777$$

$$N = 7$$

$$\sqrt[7]{3555.5} = 3.21 = 3$$

$$t_p = t_{pLH} = t_{pHL} = (N-1) \cdot 0.69 R_p (C_{01} + S(C_{TN})) + 0.69 R_p (C_{01} + C_1 / S^{N-1})$$

$$= 6 \cdot 0.69 \cdot 3.5 \cdot 10^3 (1.875 + 3 \cdot 2.8125) + 0.69 \cdot 3.5 \cdot 10^3 \left(1.875 + \frac{10 \cdot 10^{-12}}{3^6} \right)$$

$$= 1.544 \cdot 10^{-10} = 0.15 \text{ ns}$$

$$7- f = 4 \text{ GHz}$$

$$C_{IN} = 1'5 \cdot (C_{oxN} + C_{oxP}) \quad C_{OUT} = C_{oxN} + C_{oxP}$$

$$C_{IN} = 1'5 \cdot \left(\frac{25 \cdot 0'5 \cdot 0'05}{2} + \frac{25 \cdot 0'5 \cdot 0'05}{2} \right) \quad C_{OUT} = \frac{25 \cdot 0'5 \cdot 0'05}{2} + \frac{25 \cdot 0'5 \cdot 0'05}{2}$$

$$F_{osc} = \frac{1}{N \cdot (t_{PH1} + t_{PLH})}$$

$$C_L = C_{IN} + C_{OUT} = 4'6825 \cdot 10^{-15}$$

$$4 \cdot 10^9 = \frac{1}{N \cdot 2}$$

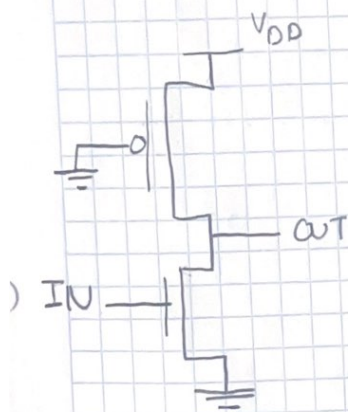
$$t_{PH1} = 0'69 \cdot 7 \cdot 10^3 \cdot (C_{IN} + C_{OUT}) = 22640'625$$

$$V = 5$$

$$8- R_n = 35 \cdot \frac{I_n = 50}{W_n = 500}$$

$$R_p = 70 \cdot \frac{I_p}{W_p = 500}$$

$$C_L = 1 \text{ pF}$$



$L_p \text{ (nm)}$	$V_{OL} \text{ (V)}$	$t_{PLH} \text{ (ns)}$	$P \text{ (}\mu\text{W)}$
50		4'23	
100		9'66	
200		19'32	

$$t_{PLH} = 0'69 R_p C_L$$

$$P = C_L V_{DD}^2 F_0$$

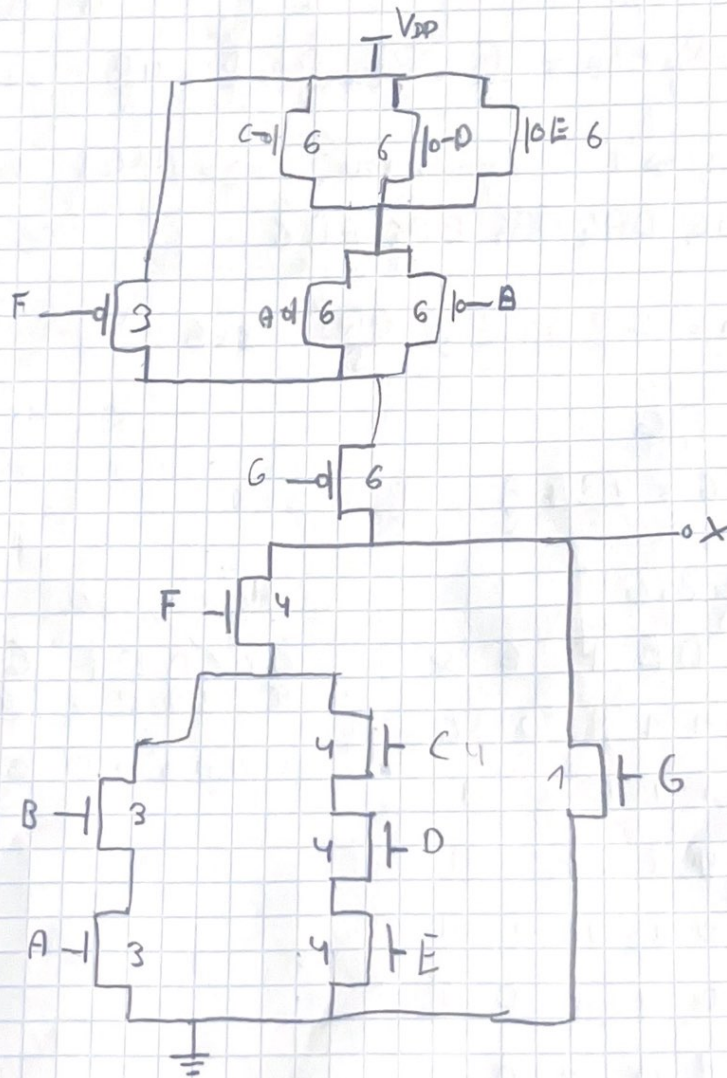
$$1- X = [(\bar{A} + \bar{B}) (\bar{C} + \bar{D} + \bar{E}) + \bar{F}] \bar{G}$$

$$X = [(\bar{A} \cdot \bar{B}) (\bar{C} \cdot \bar{D} \cdot \bar{E}) + \bar{F}] \bar{G}$$

$$X = [\overline{((A \cdot B) + (C \cdot D \cdot E)) \cdot F}] \bar{G}$$

$$X = \overline{(((A \cdot B) + (C \cdot D \cdot E)) \cdot F) + G}$$

a)



$$c) R_p \approx K_p \left(\frac{L_p}{W_p} \right) \quad R_n \approx K_n \left(\frac{L_n}{W_n} \right)$$

$$R_p = 2R_n \quad \text{PMOS} \rightarrow \text{doble ancho que NMOS}$$

$$\text{Si } s \geq 1 \quad L = L_{\min} : W = s W_{\min} \quad \frac{W}{L} = s \frac{W_{\min}}{L_{\min}}$$

$$\text{Si } s < 1 \quad W = W_{\min} \quad L = \frac{L_{\min}}{s}$$

$$s = 2 \Rightarrow \begin{cases} L_p = L_{\min} \\ W_p = 2 W_{\min} \end{cases}$$

Peor caso '1' $\rightarrow$ '0' camino más largo NMOS hasta tierra. $s_{\text{max}} = 3$

(FCDE)

$$R_F + R_A + R_B \leq R \quad \frac{R}{4} + \frac{2R}{s_{\text{max}}} \leq R$$

$$R_F + R_C + R_D + R_E = \frac{R}{s} + \frac{R}{s} + \frac{R}{s} + \frac{R}{s} = \frac{4R}{s} \quad s = 4$$

Peor caso '0' $\rightarrow$ '1' camino más largo PMOS hacia salida.
CAG, CBG, DAG, DBG, EAG, EBG

$$R_C + R_A = \frac{2R}{s} + \frac{2R + \frac{2R}{s}}{s} = \frac{6R}{s} \quad s = 6$$

Camino FG

$$R_F + R_G = \frac{2R}{s} + \frac{R}{s} = \frac{3R}{s} \quad s = 3$$

b) Peor caso

A B C D E F G X

0 0 1 1 1 1 0 0

0 1 0 1 1 1 0 1

1 0 0 1 1 1 0 1

0 1 1 0 1 1 0 1

1 0 1 0 1 1 0 1

0 1 1 1 0 1 0 1

1 0 1 1 0 1 0 1

Mejor caso

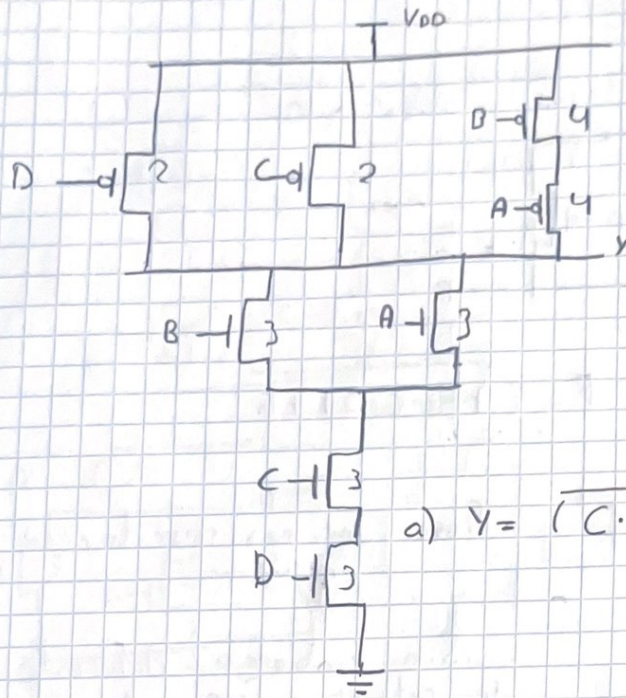
A D C D E F G X

0 1 1 1 1 1 1 0

0 0 0 0 0 0 0 1

tener en cuenta que están repetidos

2-



a) $Y = \overline{(C \cdot D) \cdot (A+B)}$

b) "1" → "0" BCD o ACD

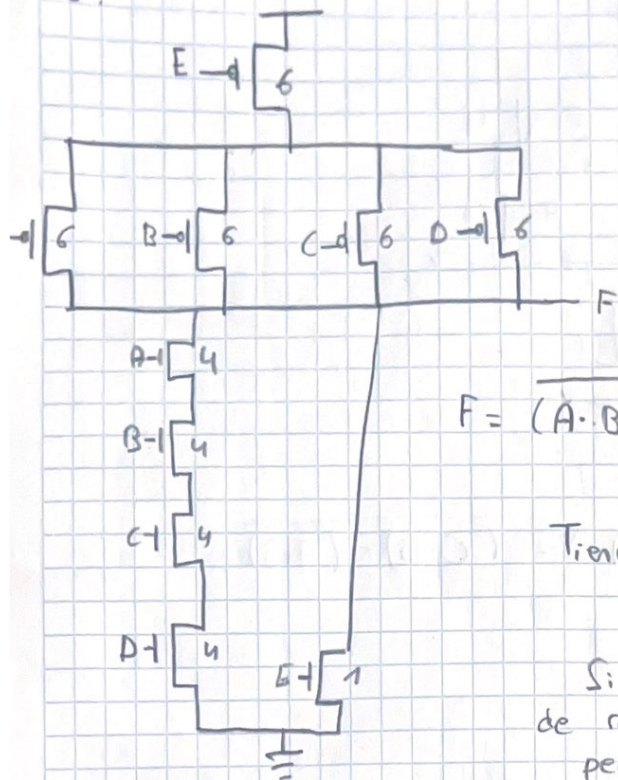
ABC D	Y
011 1	0
101 1	0
00 11	1

"0" → "1" AB

c) $R_B + R_C + R_D = \frac{R}{J} + \frac{R}{J} + \frac{R}{J} = \frac{3R}{J} \quad J=3$

$R_A + R_B = \frac{2R}{J} + \frac{2R}{J} = \frac{4R}{J} \quad J=4$

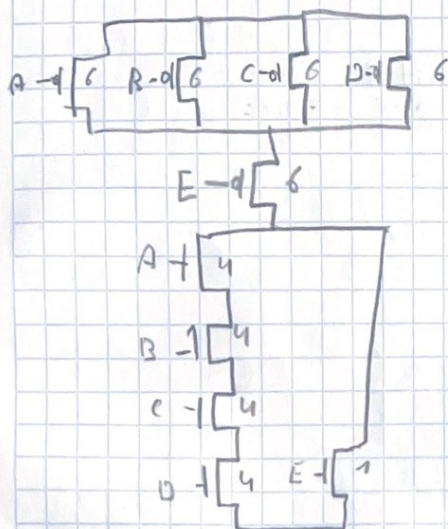
3-



$$F = \overline{(A \cdot B \cdot C \cdot D)} + E$$

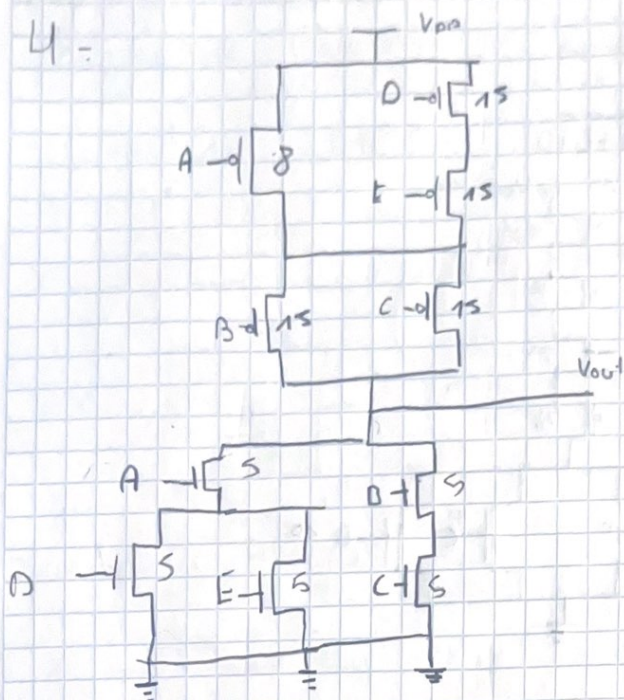
Tienen mismas resistencias equivalentes.

Sin embargo diferente tiempo de retardo porque el segundo tiene peor caso posible.



$$F = \overline{(A \cdot B \cdot C \cdot D)} + E$$

U =



$$a) F = (D+E) \cdot A + (C \cdot B)$$

$$b) t_{p2H} = t_{pH2} = 0.69 R_N C_L \Rightarrow 1 \cdot 10^{-9} = 0.69 R \cdot 1 \cdot 10^{-12}$$

$$R = 1'449 \text{ k}\Omega$$

$$R_D = 70 \frac{I_P}{V_P} \quad R_N = 35 \frac{I_N}{V_N}$$

$$R_D = 2 R_N$$

$$R_D = \frac{70 \cdot 50}{5 \cdot 500}$$

$$I_N = I_{min}$$

$$V_N = 1 \text{ V}_{min}$$

$$S_D = 5$$

$$R_N = \frac{35 \cdot 50}{5 \cdot 500}$$

$$S = 2.5$$

$$R_B + R_C = \frac{2.5 R}{5} + \frac{2.5 R}{5} = \frac{5 R}{5} = 1 = 5$$

$$R_{D,E,C} = \frac{5 R}{5} + \frac{5 R}{5} + \frac{5 R}{5} = 15$$

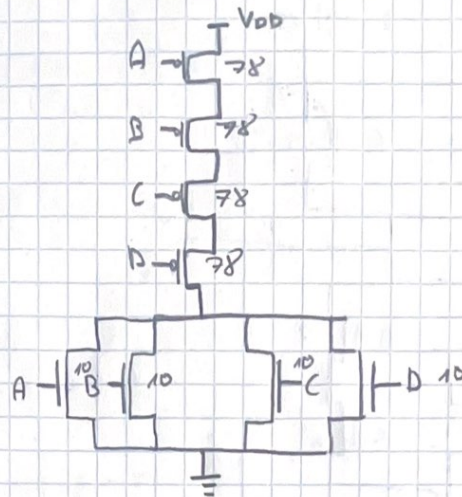
5: NOR 4 entradas operar frec superior a 1 GHz

$$G_{LC} = 1/p F_{LC}$$

A	B	C	D	S
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

$$F = \overline{A+B+C+D}$$

a)



b) $F = 1 \text{ GHz}$ $T = 1 \text{ ns}$

$$0 \rightarrow 1 \rightarrow 0$$

50% de la salida

$$t_{PHL} = t_{PLH} = 0.25 \text{ ns}$$

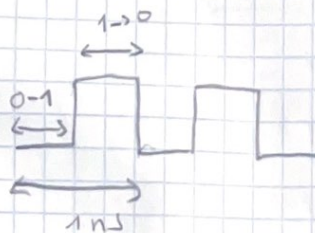
$$0.25 = 0.69 R \cdot 1 \cdot 10^{-12}$$

$$R = 362.31 \Omega$$

$$R_N = \frac{35 \cdot 50}{1 \cdot 500} = \xi = 9.66 \cdot 10^{-3}$$

$$f_p = 19.32$$

$$19.32 \cdot 4 = 77.28 \rightarrow 78$$

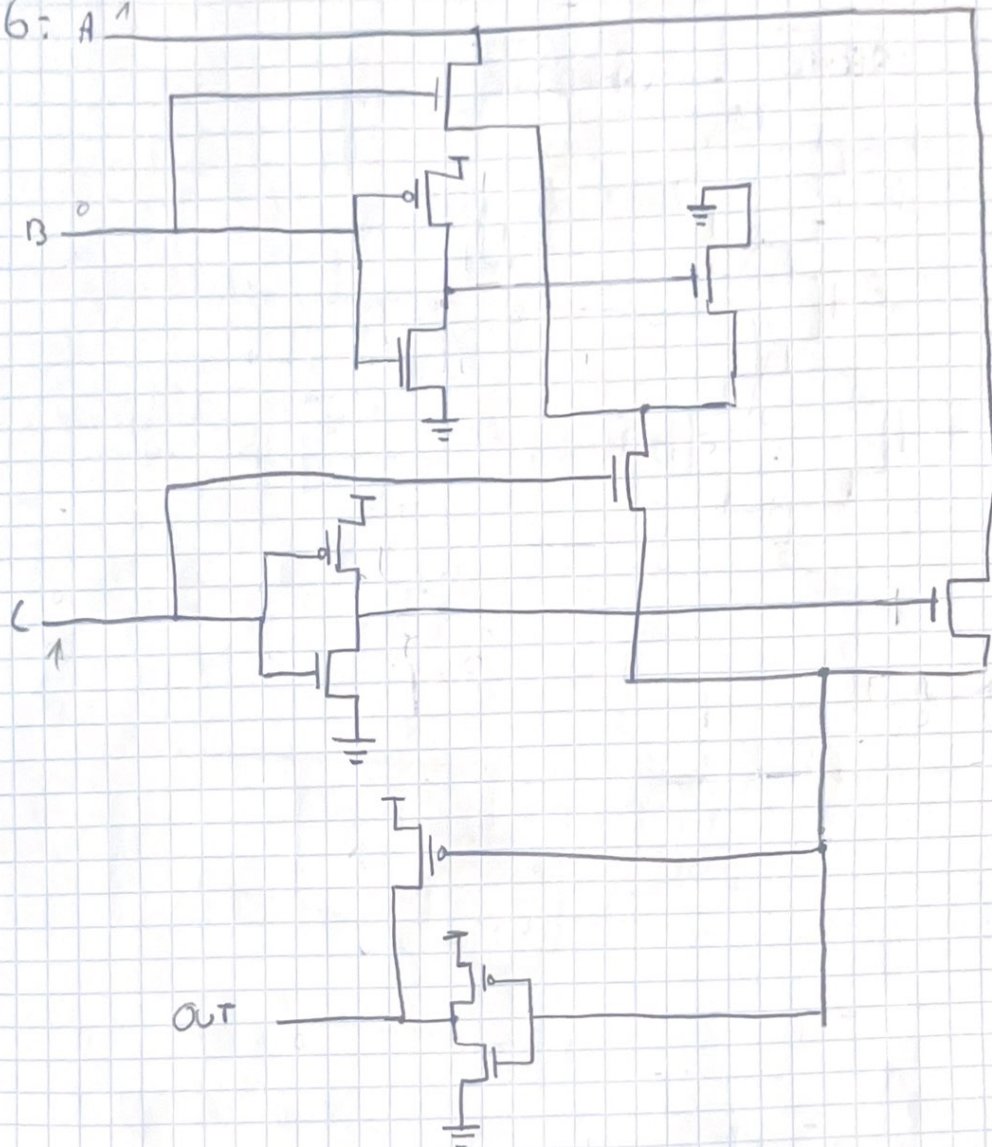


$$1 \text{ GHz} = 1 \text{ ns}$$

$$\left. \begin{array}{l} 1 \rightarrow 0 \\ 0 \rightarrow 1 \end{array} \right\} 0.5 \text{ ns}$$

$$t_{PHL} = 0.5 \text{ de llegar}$$

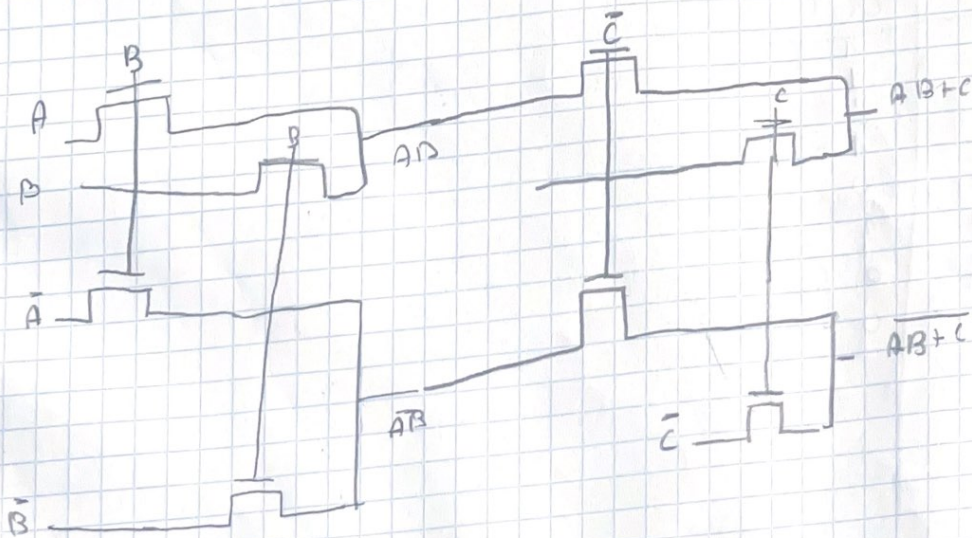
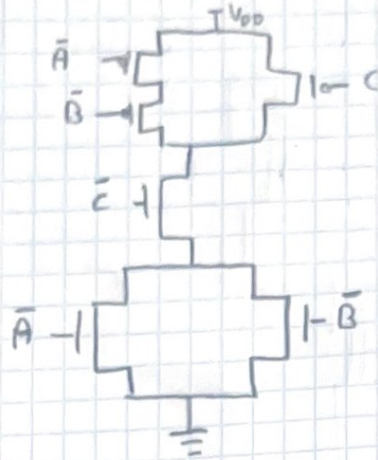
6: A¹



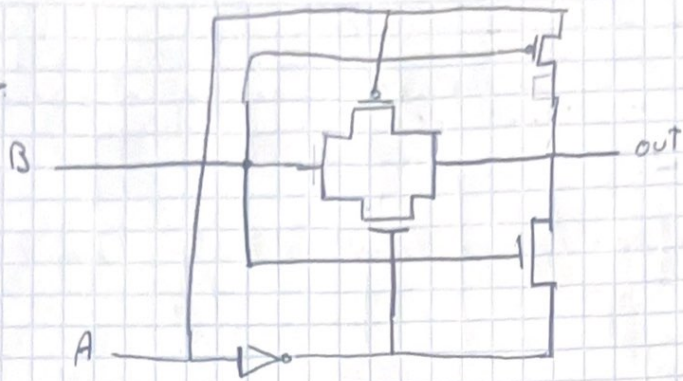
ABC	OUT
000	1
001	1
010	1
011	1
100	0
101	1
110	0
111	0

7: $F = AB + C$ con logico CPL

$$\overline{AB + C} = (\overline{A + B}) \cdot \overline{C}$$



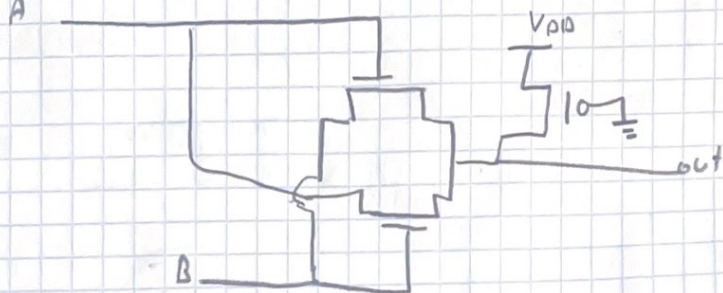
8-



A	B	OUT
0	0	0
0	1	1
1	0	1
1	1	0

EXOR

9-



a)

A	B	OUT
0	0	1
0	1	0
1	0	0
1	1	1

b)

A	B	OUT
0	0	OUT
0	1	0
1	0	0
1	1	1

$$V_{out} = \frac{V_{DD} \cdot R_{up}}{R_{up} + R_{down}}$$

